



Hazards Planning and Risk Management

LECTURES 3

Hazard and Disaster Information

Fall 2026

US – Pakistan Center for Advanced Studies in Water

The **World Disasters Report** is an annual independent publication commissioned by the International Federation of Red Cross and Red Crescent Societies (IFRC). Each year, experts present a body of research that examines challenges, trends and innovations around a specific humanitarian theme. Oct 13, 2016



World Disasters Report 2026

05/03/2026 | Document



The World Disasters Report 2026 warns harmful information as a **de facto humanitarian crisis** - undermining access to aid, eroding trust, destabilizing social cohesion and increasing risks for staff, volunteers and communities. No longer a peripheral communications issue, harmful information is an operational and strategic challenge affecting acceptance, safety and principled humanitarian action.

As the IFRC's flagship publication, the Report brings together insights from nearly 100 contributors, 60 organizations including over 30 National Red Cross and Red Crescent Societies. Grounded in experiences from across regions and community insights, it examines how harmful information, including misinformation, disinformation and dehumanizing narratives, impacts responses to disasters, public health emergencies, migration and conflict.

Across eight chapters, the Report shows why understanding harms of harmful information is critical — revealing its wide-ranging impacts and how it disrupts preparedness, response and recovery. It provides practical recommendations to strengthen information resilience, safeguard humanitarian principles and rebuild trust.

The IFRC World Disasters Report 2026 focuses on harmful information—misinformation, disinformation, hate speech, and other damaging narratives—as an emerging humanitarian challenge.

The 2026 report highlights that in modern disasters, reliable information and public trust are increasingly essential for effective, safe, and inclusive humanitarian response.

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Climate Reports

Key reports on climate impacts and solutions from around the United Nations.

What's New



**UNFCCC | NDC
Synthesis Report**

Phasing down or phasing up?
Top fossil fuel producers plan to phase out coal
despite climate promises

**UNEP | Production
Gap Report**

**Underfinanced.
Underprepared.**

UN
DEVELOPMENT
PROGRAMME

**UNEP | Adaptation
Gap Report**

Reports

The IPCC prepares comprehensive Assessment Reports about knowledge on climate change, its causes, potential impacts and response options. The IPCC also produces Special Reports, which are an assessment on a specific issue and Methodology Reports, which provide practical guidelines for the preparation of greenhouse gas inventories.

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Managing the Risks of Extreme Events and Disasters to Advance Climate Change Adaptation (SREX)



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Report by Chapters

- [Front Matter](#) [0.1MB](#)
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Chapters

- [1. Climate Change: New Dimensions in Disaster Risk, Exposure, Vulnerability, and Resilience](#) [0.8MB](#)
- [2. Determinants of Risk: Exposure and Vulnerability](#) [0.8MB](#)
- [3. Changes in Climate Extremes and their Impacts on the Natural Physical Environment](#) [19.5MB](#)
- [4. Changes in Impacts of Climate Extremes: Human Systems and Ecosystems](#) [6.6MB](#)
- [5. Managing the Risks from Climate Extremes at the Local Level](#) [1.2MB](#)
- [6. National Systems for Managing the Risks from Climate Extremes and Disasters](#) [1.2MB](#)
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REPORTS SYNTHESIS REPORT WORKING GROUPS

greenhouse gas inventories. The latest report is the [Sixth Assessment Report](#), which includes Working Group contributions and a Synthesis Report. The Working Group I contribution was released in August 2021, the [Working Group II](#) contribution in February 2022, the Working Group III contribution in April 2022 and the [Synthesis Report](#) in March 2023.

◀

Sixth Assessment Report: 2023

Synthesis Report

AR6 Synthesis Report: Climate Change 2023

March 2023

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SIXTH ASSESSMENT REPORT

FIFTH ASSESSMENT REPORT

2027 IPCC Methodology Report on Inventories for Short-lived Climate Forcers
Special Report on Climate Change and Cities
Global Warming of 1.5°C
Climate Change and Land
2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
The Ocean and Cryosphere in a Changing Climate

AR6 Synthesis Report: Climate Change 2023
AR6 Climate Change 2022: Impacts, Adaptation and Vulnerability
AR6 Climate Change 2022: Mitigation of Climate Change
AR6 Climate Change 2021: The Physical Science Basis

AR5 Synthesis Report: Climate Change 2014
AR5 Climate Change 2013: The Physical Science Basis
AR5 Climate Change 2014: Impacts, Adaptation, and Vulnerability
AR5 Climate Change 2014: Mitigation of Climate Change

VIEW ALL

Facts : UNFCCC



Name: United Nations Framework Convention on Climate Change

Adopted in the Year: 1992

Entered into Force in the Year: 1994

Headquarters: Bonn, Germany

Sector: Climate Change

Is India a Signatory: Yes, but ratified it in 1993.

The UNFCCC is not legally binding.

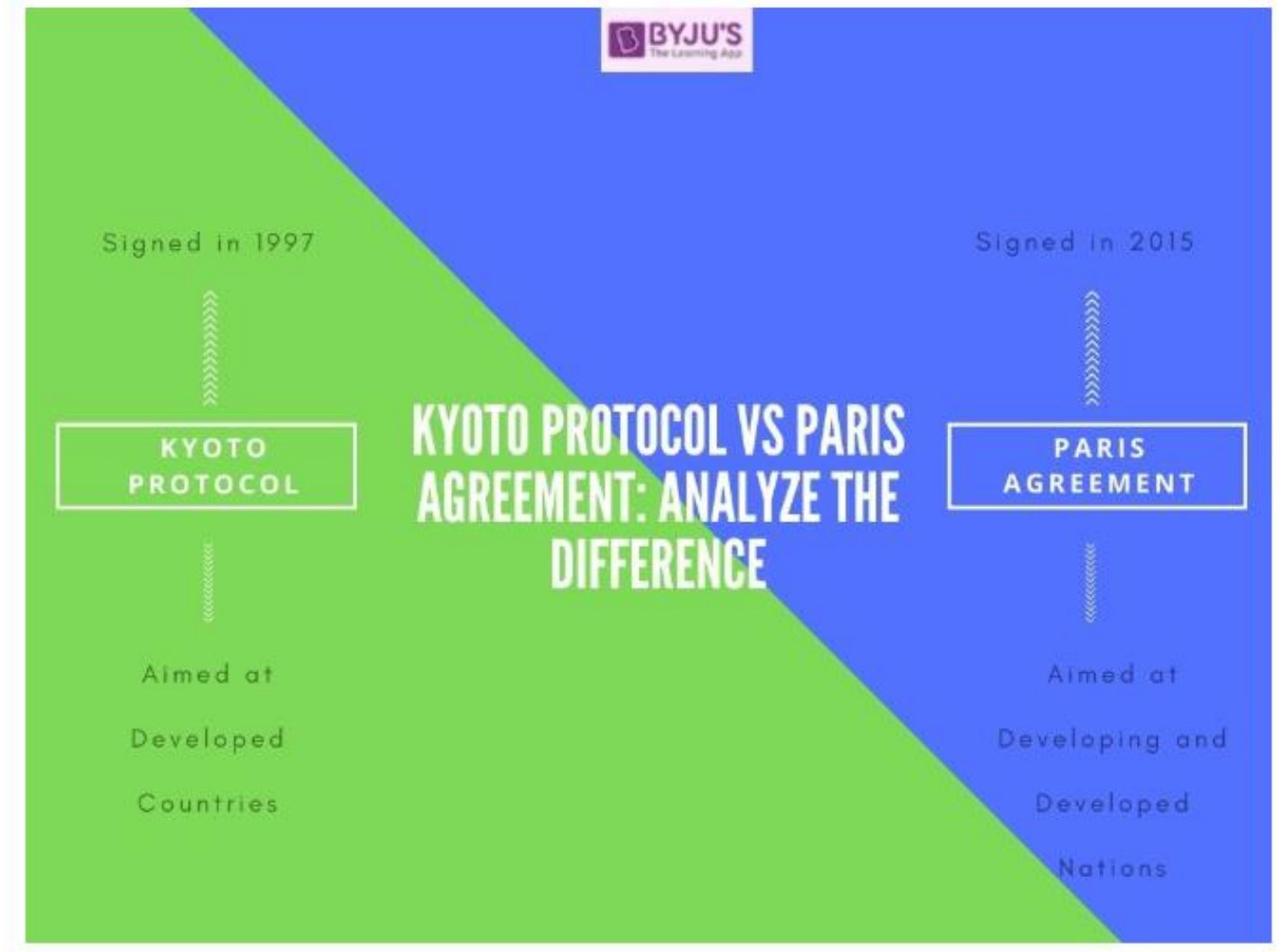
It sets a framework for limiting the emissions of greenhouse gases (GHG) by individual countries.



The UN Climate Agreements

Kyoto Protocol & Paris Agreement

Kyoto Protocol	Paris Agreement
 Was created in 1997 and ratified in 2005. Had two periods from 1997-2020.	 Signed in November of 2016. New commitments are due every 5 years
 Legally binding agreement to decrease GHG	 Not legally binding commitment to reduce emissions, increase accountability
 Original commitment to decrease overall emissions by 5% from 1990 levels	 Overall goal to limit global temperatures to 1.5 degrees celcius above pre-industrial levels
 Only required developed nations to reduce emissions	 Asked all nations to reduce emissions
 Targets are set but no determined time frame	 New set of targets declared after 5 years (these are now due in 2020)



Governing and subsidiary bodies.

The Convention, the Kyoto Protocol, and the Paris Agreement established three governing bodies: the COP for the Convention, the CMP for the Kyoto Protocol and the CMA for the Paris Agreement. The Bureau of the COP, the CMP and the CMA supports the work of the governing bodies. The UNFCCC secretariat provides organizational support and technical expertise.

Two permanent subsidiary bodies – the SBSTA and the SBI – assist the governing bodies.

The COP, the CMP and the CMA furthermore establish other ad hoc subsidiary bodies as deemed necessary to address specific issues.



Credit: UN Climate Change/ Kiara Worth

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Governing bodies

Conference of the Parties (COP)

The COP is the supreme decision-making body of the Convention. All States that are Parties to the Convention are represented at the COP, at which they review the implementation of the Convention and any other legal instruments that the COP adopts and take decisions necessary to promote the effective implementation of the Convention, including institutional and administrative arrangements.

Conference of the Parties serving as the meeting of the Parties to the Kyoto Protocol (CMP)

The Conference of the Parties, the supreme body of the Convention, shall serve as the meeting of the Parties to the Kyoto Protocol. All States that are Parties to the Kyoto Protocol are represented at the Conference of the Parties serving as the meeting of the Parties to the Kyoto Protocol (CMP), while States that are not Parties participate as observers. The CMP oversees the implementation of the Kyoto Protocol and takes decisions to promote its effective implementation.

Conference of the Parties serving as the meeting of the Parties to the Paris Agreement (CMA)

The Conference of the Parties, the supreme body of the Convention, shall serve as the meeting of the Parties to the Paris Agreement. All States that are Parties to the Paris Agreement are represented at the Conference of the Parties serving as the meeting of the Parties to the Paris Agreement (CMA), while States that are not Parties participate as observers. The CMA oversees the implementation of the Paris Agreement and takes decisions to promote its effective implementation.

BODIES

Conference of the Parties (COP).

What is the COP?

The COP is the supreme decision-making body of the Convention. All States that are Parties to the Convention are required to attend the COP, in which they review the implementation of the Convention and any other legal instruments that the COP adopts and take decisions to promote the effective implementation of the Convention, including institutional and administrative arrangements.

More Background on the COP

A key task for the COP is to review the national communications and emission inventories submitted by Parties. Based on these, the COP assesses the effects of the measures taken by Parties and the progress made in achieving the Convention's objectives.

26th Conference of Parties including countries and other stakeholders.

Events Leading up to COP26

The main developments in global environmental diplomacy leading to the situation in which the UN and member nations now find themselves are as follows.

1987 – the Montreal Protocol – a historic global environmental agreement, ratified by every country globally, to stop producing substances (CFCs) that damage the earth's protective ozone layer.

1992 – the UN Framework on Climate Change (UNFCCC) was established and ratified by 197 countries. This framework was the first global accord to address climate change specifically. It founded the annual forum – the Conference of the Parties (COP) – at which international discussions aimed at stabilizing greenhouse gas emissions occur.

2005 – the Kyoto Protocol was opened for signature in 1998, but it would not become legally binding until at least 55 countries (that, together, accounted for 50% of global carbon dioxide emissions) ratified it. In 2005, Russia became the 55th country to ratify the agreement, at which point it became legally binding. The treaty required the signatories to reduce their carbon dioxide emissions by at least 4.7% from the 1990 baseline. Carbon dioxide emissions from the signatory countries dropped by 12.5% from 1990 to 2012. However, global carbon dioxide emissions by all countries (the USA and China never signed the Kyoto protocol) have increased by 51% since the Kyoto Protocol. As a great success, the global picture [indicates](#) it has had little real

- https://wellbeingintl.org/climate-change-and-cop26/?gclid=CjwKCAiAk9itBhASEiwA1my_6wlGaJQgxgoYP4Pbdij3Gz1rib2pijUJsxCaPV3sQYWOA2nXdPiBGRoC9wYQAvD_BwE

2024 United Nations Climate Change Conference

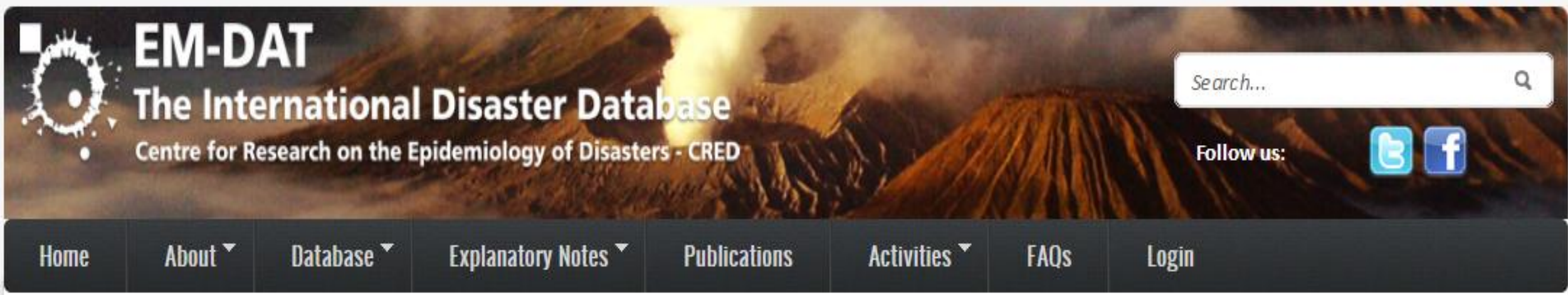


The 2024 United Nations Climate Change Conference or Conference of the Parties of the UNFCCC, more commonly known as COP29, was the 29th United Nations Climate Change... [More](#)

Source: [Wikipedia](#)

Location	Session	Conference
Dubai, United Arab Emirates	COP 28	UN Climate Change Conference - United Arab Emirates
Sharm el-Sheikh, Egypt	COP 27	Sharm el-Sheikh Climate Change Conference - November 2022
Glasgow, United Kingdom of Great Britain and Northern Ireland	COP 26	Glasgow Climate Change Conference - October-November 2021
Madrid, Spain	COP 25	UN Climate Change Conference - December 2019
Katowice, Poland	COP 24	Katowice Climate Change Conference - December 2018
Bonn, Germany	COP 23	UN Climate Change Conference - November 2017
Marrakech, Morocco	COP 22	Marrakech Climate Change Conference - November 2016
Paris, France	COP 21	Paris Climate Change Conference - November 2015
Lima, Peru	COP 20	Lima Climate Change Conference - December 2014
Warsaw, Poland	COP 19	Warsaw Climate Change Conference - November 2013
Doha, Qatar	COP 18	Doha Climate Change Conference - November 2012
Durban, South Africa	COP 17	Durban Climate Change Conference - November 2011
Cancun, Mexico	COP 16	Cancun Climate Change Conference - November 2010
Copenhagen, Denmark	COP 15	Copenhagen Climate Change Conference - December 2009
Poznan	COP 14	Poznan Climate Change Conference -

The next UN Climate Change Conference, COP31, will take place in Antalya, Türkiye, from November 9 to 20, 2026.



EM-DAT: The International Disaster Database

Centre for Research on the Epidemiology of
Disasters (CRED)

EM-DAT

- EM-DAT provides **geographical, temporal, human and economic** information on disasters at the country level.

<https://www.emdat.be/>
<https://public.emdat.be/>

EM-DAT Screenshots

Examples of database: Profiles and Trends



Geophysical

Earthquake

Mass
Movement
(dry)

Volcanic
activity



Hydrological

Flood

Landslide

Wave
action



Meteorological

Storm

Extreme
temperature

Fog



Climatological

Drought

Glacial lake
outburst

Wildfire



Biological

Animal
accident

Epidemic

Insect
infestation



Extra-terrestrial

Impact

Space
weather

General Classification

Disaster Group	Disaster Subgroup	Definition	Disaster Main Type
Natural	Geophysical	A hazard originating from solid earth. This term is used interchangeably with the term geological hazard.	Earthquake
			Mass Movement
			Volcanic activity
	Meteorological	A hazard caused by short-lived, micro- to meso-scale extreme weather and atmospheric conditions that last from minutes to days.	Extreme Temperature
			Fog
			Storm
	Hydrological	A hazard caused by the occurrence, movement, and distribution of surface and subsurface freshwater and saltwater.	Flood
			Landslide
			Wave action
	Climatological	A hazard caused by long-lived, meso- to macro-scale atmospheric processes ranging from intra-seasonal to multi-decadal climate variability.	Drought
			Glacial Lake Outburst
			Wildfire
	Biological	A hazard caused by the exposure to living organisms and their toxic substances (e.g. venom, mold) or vector-borne diseases that they may carry. Examples are venomous wildlife and insects, poisonous plants, and mosquitoes carrying disease-causing agents such as parasites, bacteria, or viruses (e.g. malaria).	Epidemic
			Insect infestation
			Animal Accident
	Extraterrestrial	A hazard caused by asteroids, meteoroids, and comets as they pass near-earth, enter the Earth's atmosphere, and/or strike the Earth, and by changes in interplanetary conditions that effect the Earth's magnetosphere, ionosphere, and thermosphere.	Impact
			Space weather

Technological	Industrial accident		Chemical spill
			Collapse
			Explosion
			Fire
			Gas leak
			Poisoning
			Radiation
			Other
	Transport accident		Air
			Road
			Rail
			Water
	Miscellaneous accident		Collapse
			Explosion
			Fire
			Other

Glossary

- <http://www.emdat.be/Glossary>

Disaster Profiles

(Last data update: 19/07/2016)

This section provides a "pre-designed" summary and profile of disasters. The profile includes a summary of events from 1900, as well as the top 10 disasters.

If you need more detailed information, please use the [Advanced search](#) or the [Disaster list](#) options.

Search criteria

Period

From: 2010 To: 2016

☒ Natural disasters
☐ Technological disasters

Disaster type:

Disaster subgroup: Hydrological

Display disaster profiles

TOP Disasters (Show maximum 10 disasters)

Total deaths

Total Affected

Total Economic Damage

Save table as CSV file

Disaster No.	Country	Disaster type	Date
2013-0192	India	Flood	12-06-2013
2010-0341	Pakistan	Flood	28-07-2010
2010-0377	China	Landslide	07-08-2010
2010-0239	China	Flood	29-05-2010
2011-0014	Brazil	Flood	11-01-2011
2011-0326	Thailand	Flood	05-08-2011
2011-0347	Pakistan	Flood	12-08-2011
2012-0363	Pakistan	Flood	00-08-2012
2011-0227	China	Flood	01-06-2011
2014-0134	Afghanistan	Flood	24-04-2014

Summary

Save table as CSV file

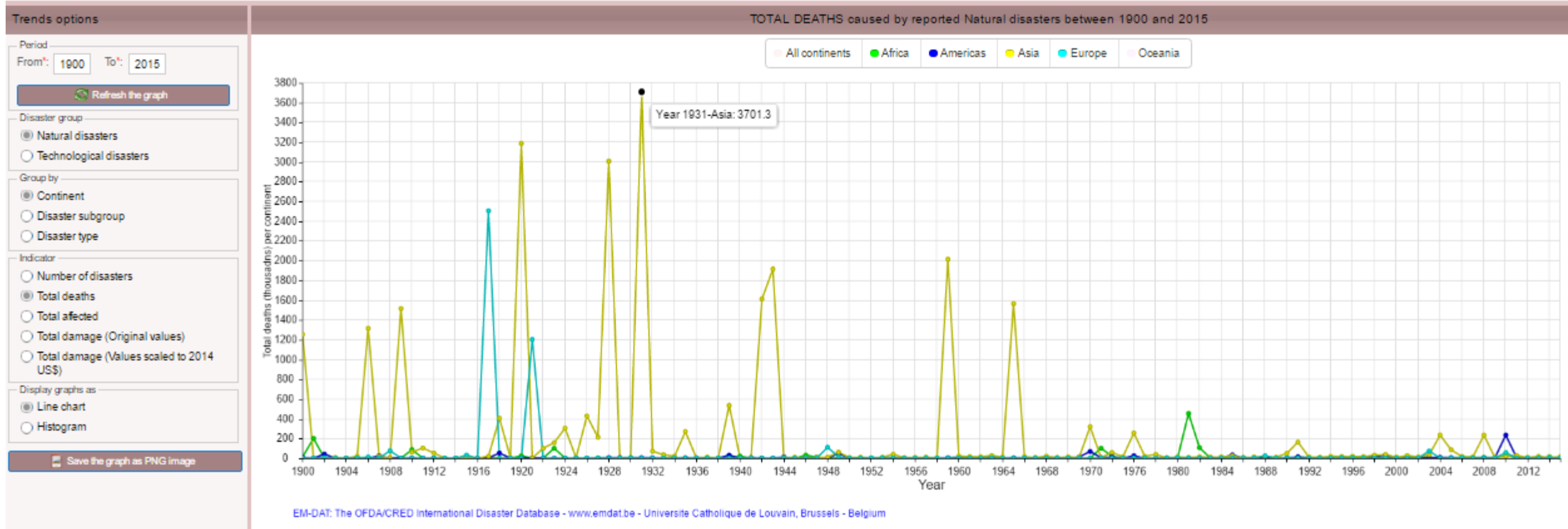
Continent	Disaster subtype	Events count	Total_deaths	Total affected	Damage ('000)
Continent: Africa					
	--	23	351	1530627	25000
	Flash flood	22	647	205249	64000
	Landslide	13	601	29848	0
	Riverine flood	178	3775	19544224	2719815
Continent: Americas					
	--	18	132	419993	2110000
	Coastal flood	1	0	199000	0
	Flash flood	17	335	354337	4519000
	Landslide	18	1051	69085	725000
	Riverine flood	168	3874	14419849	26230515
Continent: Asia					
	--	50	2127	9633333	9502200
	Avalanche	13	801	36916	3000

Disaster Trends

(Last data update: 19/07/2016)

USE HINTS:

- Show / Hide graphs by selecting / unselecting from legend items above the graphs.
- Move the mouse over the small circles on a line chart or the bars of a histogram to see the values of the selected indicator.
- Charts are updated automatically everytime you change an option **EXCEPT** for period limits where you have to click "Refresh the graph" button to update.
- Values for the indicator Total damage (Values scaled to 2014 US\$) are **ONLY** available for years from 1960 to 2014.



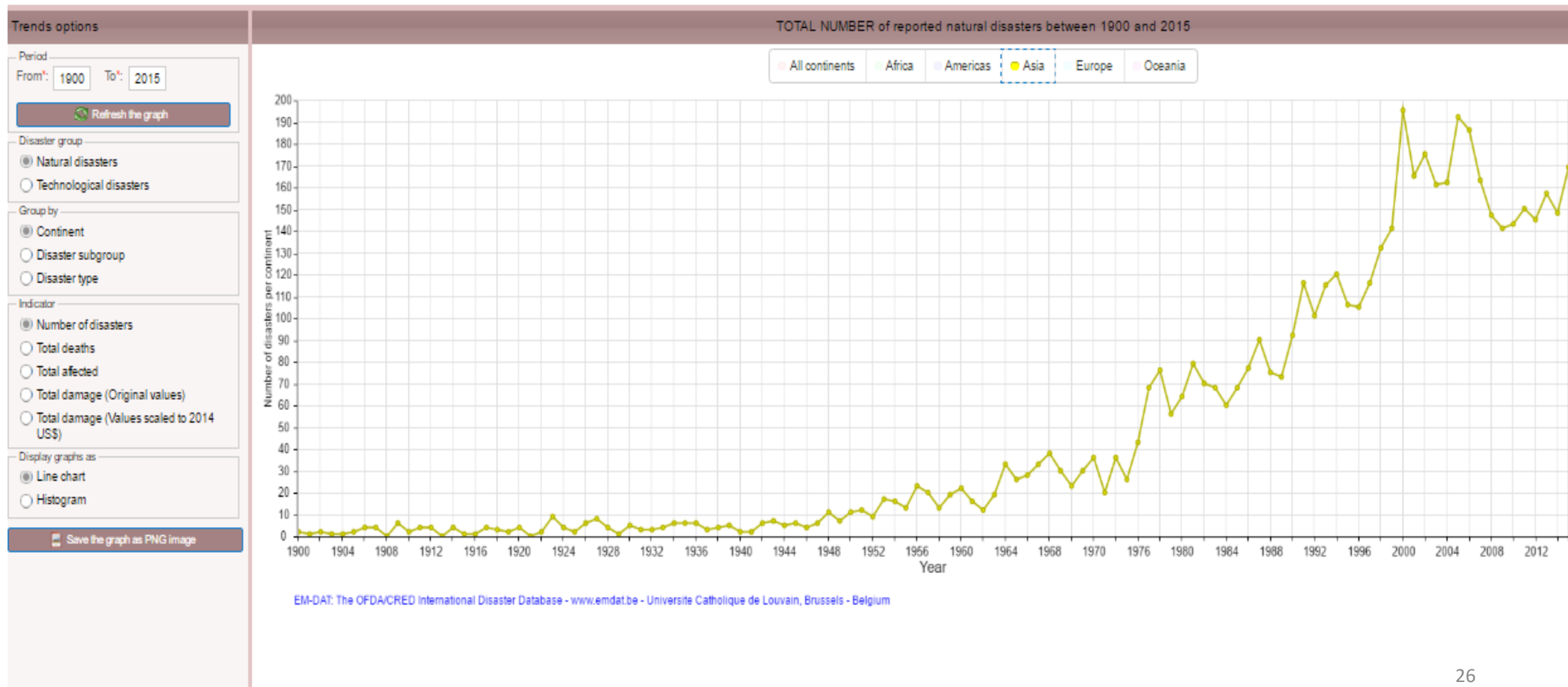
Any comment on these trends?

Disaster Trends

(Last data update: 19/07/2018)

USE HINTS:

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Disaster Trends

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Trends options

Period

From: 1900 To: 2015

Refresh the graph

Disaster group

☐ Natural disasters

☒ Technological disasters

Group by

☒ Continent

☐ Disaster subgroup

☐ Disaster type

Indicator

☐ Number of disasters

☒ Total deaths

☐ Total affected

☐ Total damage (Original values)

☐ Total damage (Values scaled to 2014 US\$)

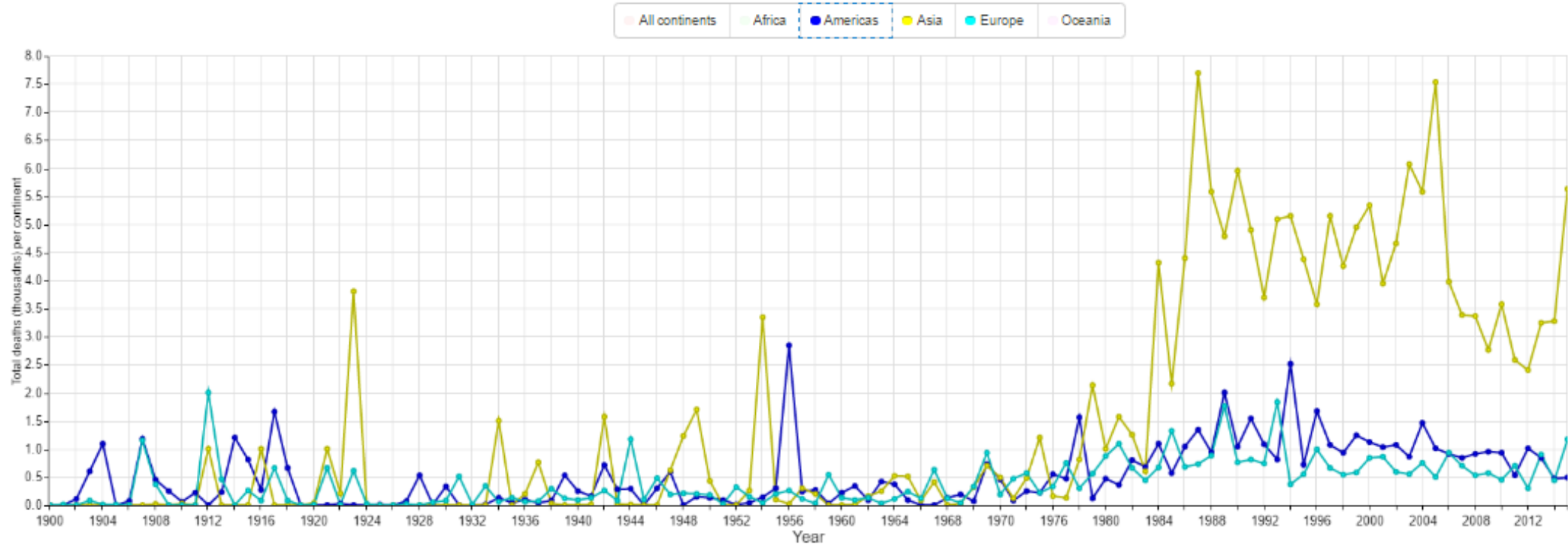
Display graphs as

☒ Line chart

☐ Histogram

Save the graph as PNG image

TOTAL DEATHS caused by reported Technological disasters between 1900 and 2015



EM-DAT: The OFDA/CRED International Disaster Database - www.emdat.be - Université Catholique de Louvain, Brussels - Belgium

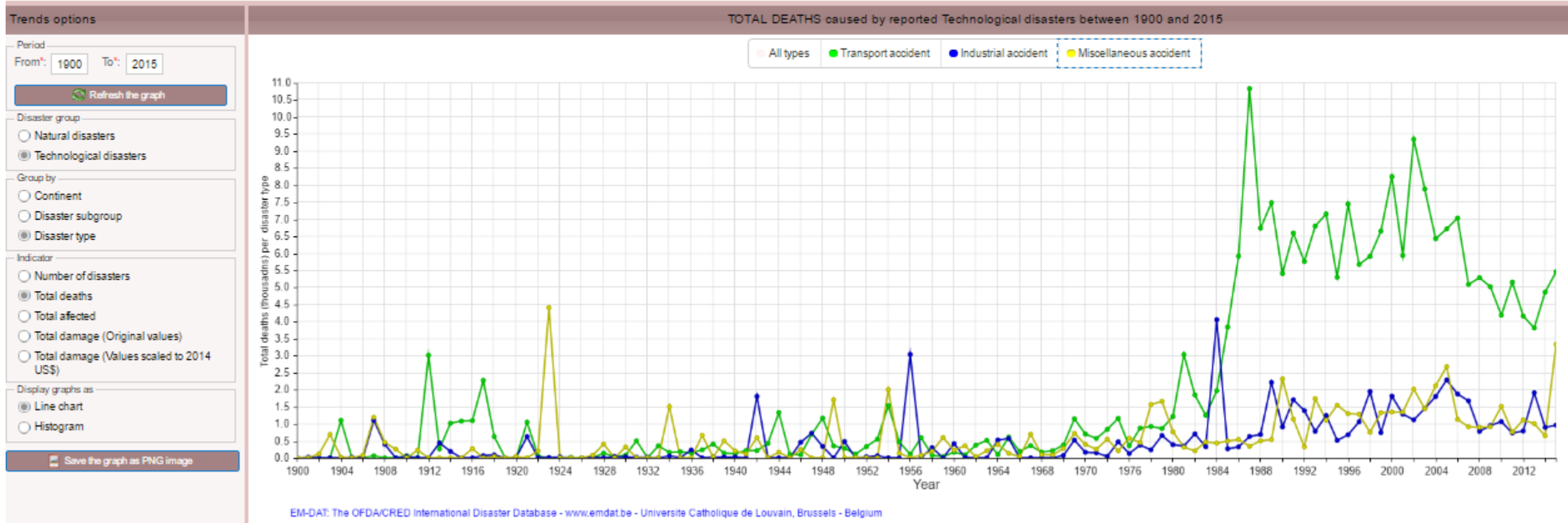
Any comment on these trends?

Disaster Trends

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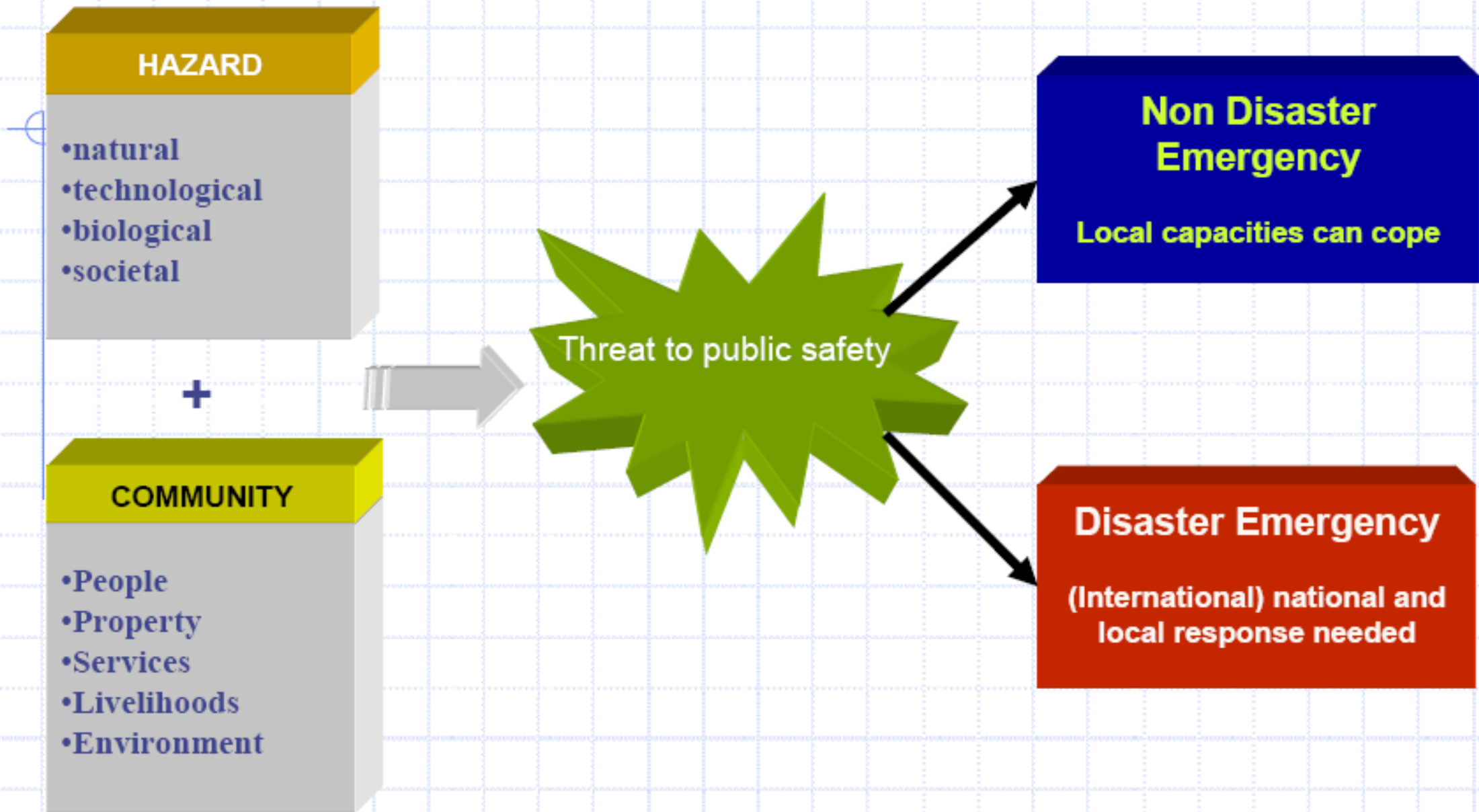
EM-DAT Criteria

- EM-DAT includes all disasters from 1900 until the present, conforming to at least one of the following criteria:
 - 10 or more people reported killed
 - 100 people or more reported affected
 - declaration of a state of emergency
 - call for international assistance.

How does Risk Develop?

- Hazard is a danger that a disaster event may develop.
- A hazard may cause a disaster to a vulnerable society.
- A disaster is a function of the risk process.
- Risk results from the combination of
 - exposure to a hazard,
 - conditions of vulnerability and
 - insufficient capacity or measures to reduce the potential negative consequences of risk.

- A disaster happens when a hazard impacts on the vulnerable population and causes damages, casualties and disruptions.
- A hazard is disastrous only when it affects people, their properties and activities.
- For example; an earthquake in an uninhabited desert cannot be considered a disaster, no matter how strong the intensities produced
- Disaster occurs only when *hazards* and *vulnerability* meet



Reading Assignment

1. Explore further at <http://www.emdat.be/database>

2. Look at

http://unisdr.org/files/7817_UNISDRTerminologyEnglish.pdf

For subject related terminologies

Assignment

Go to the EM-DAT database and determine
Historically, which disaster type has more

1. Total number?
2. Total Deaths?
3. Total affected?
4. Total estimated damages?

You will find out that the answers to 1, 2, and 3 would not be the same.

Do you have any explanation for this difference?

Discussion/Comment/Question

